HDOC Day-42 Activity

Menu-Driven C Program - String Operations Using Separate Functions

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Aim

To write a menu-driven C program using separate user-defined functions to remove vowels from a string, find the first repeating lowercase alphabet, and check whether two strings are anagrams of each other.

Header Files Used

stdio.h for input/output, **string.h** for string handling, and **ctype.h** for character conversion and checking.

Operations

1. Remove all vowels from a string.
2. Find the first repeating lowercase alphabet in a string.
3. Check whether two strings are anagrams of each other.
4. Exit.

Algorithm

1. Display the menu repeatedly using a do-while loop.
2. Read the user choice.
3. For choice 1, read a string and copy only non-vowel characters into another string.
4. For choice 2, read a string and use a frequency array to find the first lowercase letter that repeats.
5. For choice 3, read two strings, ignore spaces and letter case, count character frequencies, and compare them.
6. For choice 4, terminate the program.
7. Display an error message for an invalid menu choice.

C Program

```
#include <stdio.h>
#include <string.h>
#include <ctype.h>

int isVowel(char ch)
{
    ch = tolower((unsigned char)ch);
    return ch == 'a' || ch == 'e' || ch == 'i' ||
           ch == 'o' || ch == 'u';
}

void removeVowels(const char str[], char result[])
{
    int i, j = 0;

    for (i = 0; str[i] != '\0'; i++)
    {
        if (!isVowel(str[i]))
            result[j++] = str[i];
    }
    result[j] = '\0';
}
char firstRepeatingLowercase(const char str[])
```

```
{
    int freq[26] = {0};
    int i;

    for (i = 0; str[i] != '\0'; i++)
    {
        if (str[i] >= 'a' && str[i] <= 'z')
            freq[str[i] - 'a']++;
    }

    for (i = 0; str[i] != '\0'; i++)
    {
        if (str[i] >= 'a' && str[i] <= 'z' &&
            freq[str[i] - 'a'] > 1)
            return str[i];
    }

    return '\0';
}

int areAnagrams(const char str1[], const char str2[])
{
    int freq[256] = {0};
    int i;
    unsigned char ch;

    for (i = 0; str1[i] != '\0'; i++)
    {
        if (!isspace((unsigned char)str1[i]))
        {
            ch = (unsigned char)tolower((unsigned char)str1[i]);
            freq[ch]++;
        }
    }

    for (i = 0; str2[i] != '\0'; i++)
    {
        if (!isspace((unsigned char)str2[i]))
        {
            ch = (unsigned char)tolower((unsigned char)str2[i]);
            freq[ch]--;
        }
    }

    for (i = 0; i < 256; i++)
    {
        if (freq[i] != 0)
            return 0;
    }

    return 1;
}

int main()
{
    int choice;
    char str1[200], str2[200], result[200];
    char repeated;

    do
    {
        printf("\n----- MENU -----\n");
        printf("1. Remove all vowels from a string\n");
        printf("2. Find first repeating lowercase alphabet\n");
        printf("3. Check whether two strings are anagrams\n");
        printf("4. Exit\n");
        printf("Enter your choice: ");
        scanf("%d", &choice);
        getchar();

        switch (choice)
        {
            case 1:
                printf("Enter a string: ");
                fgets(str1, sizeof(str1), stdin);
                str1[strcspn(str1, "\n")] = '\0';

                removeVowels(str1, result);
                printf("String after removing vowels: %s\n", result);
                break;

            case 2:
```

```
                printf("Enter a string: ");
                fgets(str1, sizeof(str1), stdin);
                str1[strcspn(str1, "\n")] = '\0';

                repeated = firstRepeatingLowercase(str1);
                if (repeated != '\0')
                    printf("First repeating lowercase alphabet: %c\n",
                           repeated);
                else
                    printf("No repeating lowercase alphabet found.\n");
                break;

            case 3:
                printf("Enter first string: ");
                fgets(str1, sizeof(str1), stdin);
                str1[strcspn(str1, "\n")] = '\0';

                printf("Enter second string: ");
                fgets(str2, sizeof(str2), stdin);
                str2[strcspn(str2, "\n")] = '\0';

                if (areAnagrams(str1, str2))
                    printf("The strings are anagrams.\n");
                else
                    printf("The strings are not anagrams.\n");
                break;

            case 4:
                printf("Exiting program...\n");
                break;

            default:
                printf("Invalid choice! Please try again.\n");
        }

    } while (choice != 4);

    return 0;
}
```

Sample Output

```
----- MENU -----
1. Remove all vowels from a string
2. Find first repeating lowercase alphabet
3. Check whether two strings are anagrams
4. Exit
Enter your choice: 1
Enter a string: Gurveer Singh Cheema
String after removing vowels: Grvr Sngh Chm

Enter your choice: 2
Enter a string: programming
First repeating lowercase alphabet: r

Enter your choice: 3
Enter first string: listen
Enter second string: silent
The strings are anagrams.

Enter your choice: 4
Exiting program...
```

Function Explanation

isVowel() converts a character to lowercase and checks whether it is a, e, i, o, or u.

removeVowels() scans the original string and copies only non-vowel characters to the result string.

firstRepeatingLowercase() first counts the frequency of lowercase letters, then scans the string again. The first lowercase character whose frequency is greater than one is returned.

areAnagrams() uses a frequency array. Characters of the first string increase the frequency and characters of the second string decrease it. If every frequency becomes zero, the strings are anagrams. Spaces and differences in letter case are ignored.

Important Concepts

Vowel removal: Keep every character except a, e, i, o and u.
Repeating character: A character occurring more than once. The answer must be the first such lowercase character in the original string.
Anagram: Two strings are anagrams when they contain the same characters with the same frequencies in a different or same order, for example **listen** and **silent**.

Result

Thus, the menu-driven C program was implemented successfully using separate functions for vowel removal, detection of the first repeating lowercase alphabet, and anagram checking.

Quick Viva Questions

1. What is an anagram? Two words or strings containing the same characters with the same frequencies.
2. Why is tolower() used? It converts uppercase letters to lowercase so case differences can be ignored where required.
3. Why is a frequency array useful? It stores how many times each character occurs.
4. What does isspace() check? It checks whether a character is whitespace.
5. Why is strcspn() used after fgets()? It helps remove the newline character stored by fgets().
6. Which loop keeps the menu running? A do-while loop.
7. What is returned when no repeating lowercase alphabet exists? The null character **\0**.